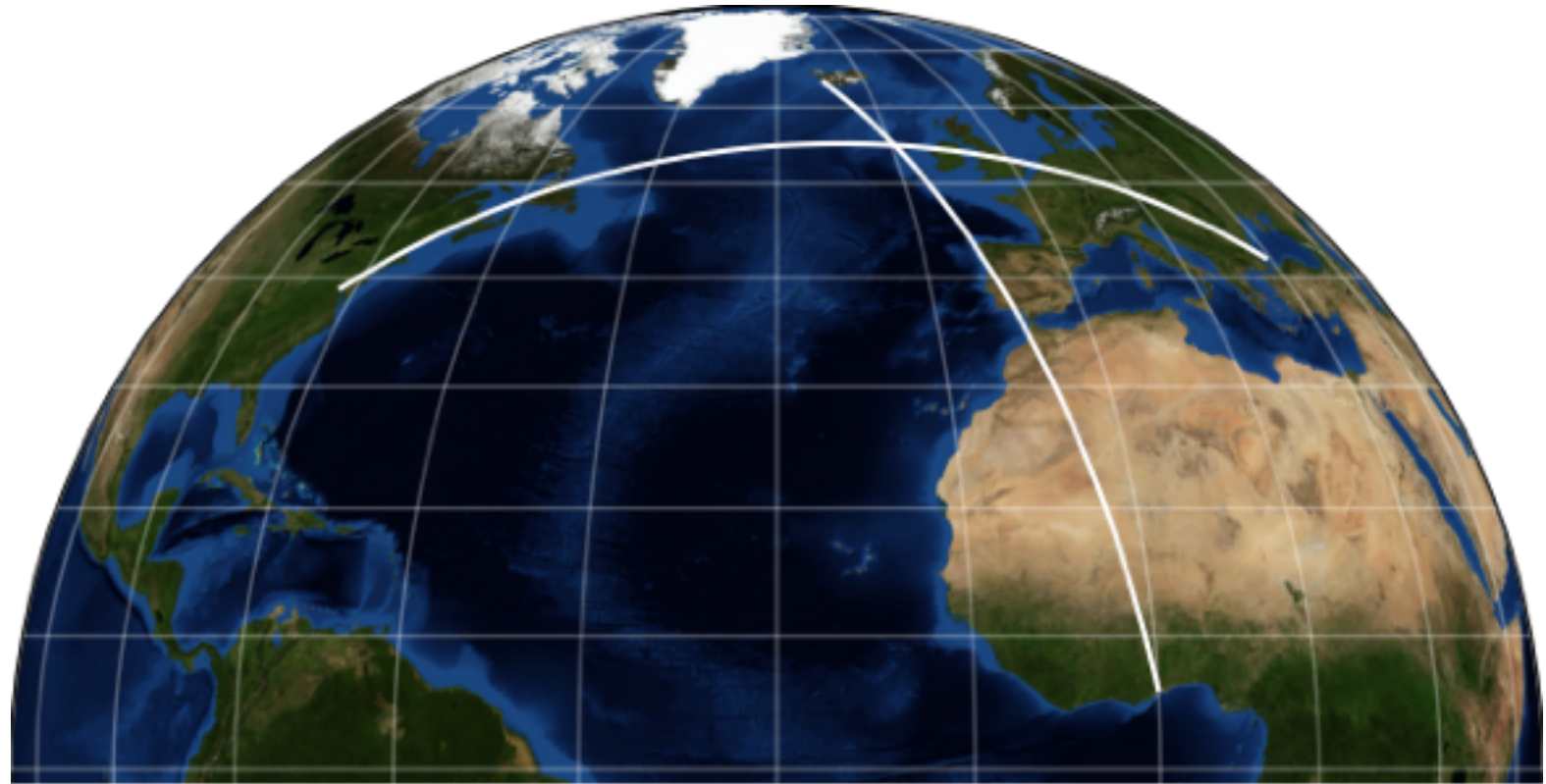


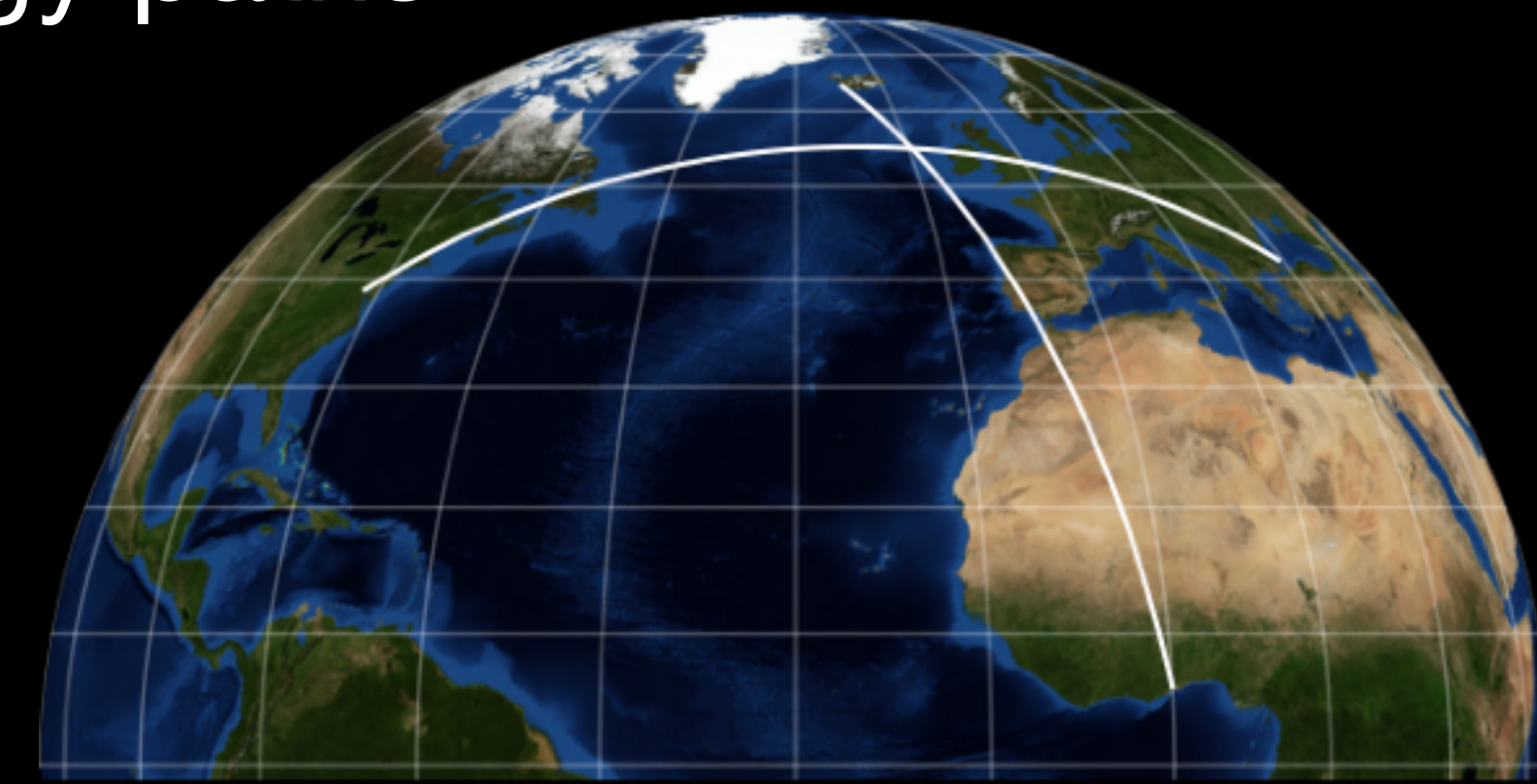
geodesic distance



OT as Geodesics

- Benamou–Brenier shows W_2 is a **geodesic distance** in $\mathcal{P}(\mathcal{X})$
- Geodesic: shortest constant speed path connecting two points in a space
- Geodesic distance: one where distances = lengths of shortest paths
- Interpolation = evolving densities along minimal-energy paths

Geodesics in a sphere:



Displacement Interpolation